Geometry-dependent electric field noise in trapped-ion systems: a noise-recycling interpretation of open and enclosed electrode geometries

Ayush Nadiger[1]

[1] *Department of Electrical and Computer Engineering*
Department of Mathematics and Statistics,
University of Massachusetts Amherst, Amherst, Massachusetts 01003, USA

(Dated: April 5, 2026)

The heating rate of a trapped ion due to fluctuating patch potentials on electrode surfaces factorizes into a surface noise spectrum $S_{\rm patch}(\omega)$ and a geometry functional $\mathcal{G}_j(\Omega; r_0)$ that is computable from the trap design. We compute $\mathcal{G}_j$ numerically for three 2D trap families—surface-electrode slots, blade traps, and two-plate enclosed traps—and one axisymmetric stylus geometry, using a Dirichlet Green's function decomposition solved with finite elements. We observe systematic trends that are naturally organized by a noise-recycling picture, in which reflecting electrode surfaces retain electrostatic image paths that would otherwise escape through openings, thereby enhancing the field noise at the ion. Three principal results emerge: (i) in the 2D cross-section, enclosing the ion between two electrode plates amplifies $\mathcal{G}_y$ by a factor of ~ 2.5 relative to an open surface trap at matched ion–electrode distance; (ii) the blade angle in a four-blade trap controls the heating anisotropy $\mathcal{G}_y/\mathcal{G}_x$ over a range of 0.5–3.9; (iii) moving the ion within a fixed asymmetric geometry produces deviations from the d^{-4} scaling law, with an effective exponent of -4.7 in a stylus trap. The qualitative trends are expected to be robust, though their magnitudes may be reduced in full 3D devices; together they motivate targeted heating-rate and anisotropy measurements in current trap architectures.

I. INTRODUCTION

A trapped ion in a radiofrequency Paul trap experiences anomalous motional heating at a rate [1, 2]

$$\dot{\bar{n}}_j = \frac{q^2}{4m\hbar\omega_j} S_{E_j}(\omega_j), \tag{1}$$

where ω_j is the secular frequency and S_{E_j} is the power spectral density of electric field noise at the ion position r_0. Within the patch-potential model [1–3], the spectral density factorizes as

$$S_{E_j}(\omega) = S_{\rm patch}(\omega)\, \mathcal{G}_j(\Omega; r_0), \tag{2}$$

separating the material-dependent surface noise spectrum $S_{\rm patch}$ from the deterministic geometry functional $\mathcal{G}_j$, which encodes the electrostatic coupling between electrode surface fluctuations and the field at the ion. The geometry functional is [3]

$$\mathcal{G}_j(\Omega; r_0) = \int_{\Gamma_e} \left|K_j(r'; \Omega, r_0)\right|^2 dA', \tag{3}$$

where $K_j(r') = \nabla_{r_0}^{(j)}[\partial G/\partial n']$ is the geometric kernel, $G(r_0, r')$ is the Dirichlet Green's function for the Laplacian on the vacuum domain Ω, and Γ_e is the electrode boundary.

Low, Herskind, and Chuang [3] computed $\mathcal{G}_j$ analytically for several symmetric geometries (sphere, disk, spheroid) and established that $\mathcal{G}_j$ scales as d^{-4} under uniform scaling of the domain. Lin, Low, and Chuang [4] extended this to rough surfaces. Safavi-Naini *et al.* [11] developed microscopic noise models connecting surface adsorbate dynamics to the observed spectral densities, while several experimental groups have directly measured the distance dependence of anomalous heating [6–8]. Surface treatment studies [9] and trap design analyses [10] have further demonstrated that both material preparation and electrode geometry influence heating rates; see Ref. [12] for a comprehensive review. Here we focus on trap geometries with gaps, openings, or partial enclosure—including surface-electrode traps, blade traps, two-plate traps, and stylus traps—where image paths can either escape through openings or be retained by reflecting surfaces.

An important distinction, which we emphasize throughout, is between *uniform* scaling (the entire domain scales with d, giving the exact d^{-4} law) and *ion-position sweeps* within a fixed geometry, where the effective exponent depends on the local geometric structure visible from the ion. The stylus trap provides the clearest illustration of this distinction.

We compute $\mathcal{G}_j$ for four geometry families (Fig. 1), introduce a geometric interpretation (*noise recycling*) in which reflecting boundaries retain image paths that would otherwise escape, and identify three experimentally accessible consequences of this picture.

II. METHOD

A. Geometry functional and relative heating predictions

The central theoretical object is the geometry functional $\mathcal{G}_j$ entering Eq. (2). Once $\mathcal{G}_j$ is known, Eq. (1) implies a relative-prediction rule for heating rates. For

two traps or two motional modes labeled (1) and (2),

$$\frac{\dot{\bar{n}}_j^{(1)}}{\dot{\bar{n}}_j^{(2)}} = \frac{\omega_j^{(2)}}{\omega_j^{(1)}} \frac{S_{\rm patch}(\omega_j^{(1)})}{S_{\rm patch}(\omega_j^{(2)})} \frac{\mathcal{G}_j^{(1)}}{\mathcal{G}_j^{(2)}}. \tag{4}$$

In the matched-frequency case with comparable surface preparation,

$$\frac{\dot{\bar{n}}_j^{(1)}}{\dot{\bar{n}}_j^{(2)}} = \frac{\mathcal{G}_j^{(1)}}{\mathcal{G}_j^{(2)}}. \tag{5}$$

The primary role of the numerics below is therefore to compute *relative* geometry-induced changes in heating. Three specific experimental protocols follow from this: (a) comparing heating rates in traps with different enclosure at matched d and comparable surface preparation, with the frequency dependence in Eq. (4) either matched or corrected for, tests Prediction 1; (b) measuring two radial heating rates in a blade trap with known θ tests Prediction 2; (c) fitting $\mathcal{G}_j$ vs. ion position in a fixed asymmetric trap tests Prediction 3.

B. Green's function decomposition

We work primarily in two spatial dimensions. This is exact for trap cross-sections with translational invariance along the trap axis (as in idealized blade and two-plate/multi-wafer geometries), which are well-described by their 2D cross-section. The stylus trap is treated in the full axisymmetric (effectively 3D) formulation.

The Dirichlet Green's function is decomposed as

$$G(r, r') = G_0(r, r') + H(r, r'), \tag{6}$$

where $G_0 = -(2\pi)^{-1} \ln|r - r'|$ is the free-space 2D Green's function and H is the regular part satisfying

$$-\Delta H = 0 \quad \text{in } \Omega, \qquad H\big|_{\partial\Omega} = -G_0\big|_{\partial\Omega}. \tag{7}$$

Since $r' \in \Omega$ (interior), $G_0|_{\partial\Omega}$ is smooth and the Dirichlet problem for H is well-posed.

For the axisymmetric stylus, the decomposition uses the 3D free-space Green's function $G_0 = -(4\pi|r - r'|)^{-1}$ and the bilinear form $a(u, v) = \int (\nabla u \cdot \nabla v)\, r\, dr\, dz$, with $\mathcal{G}_j = 2\pi \int |K_j|^2 r'\, dl'$.

C. Numerical pipeline

For each geometry: (1) Gmsh generates a triangular mesh with adaptive refinement near electrode edges ($\sim 0.3\,\mu$m) and the ion position ($\sim 0.5\,\mu$m). (2) H is solved using P2 Lagrange elements and a direct (LU) solver in DOLFINx [5]. (3) $\partial G/\partial n = \partial G_0/\partial n + \partial H/\partial n$ is evaluated at boundary facet midpoints. (4) The kernel K_j is obtained by finite-differencing the ion position: $K_j(r') \approx [\partial G/\partial n(r'; r_0 + \delta\hat{e}_j) - \partial G/\partial n(r'; r_0 - \delta\hat{e}_j)]/(2\delta)$ with $\delta = 0.005\, d$. (5) $\mathcal{G}_j$ is computed by midpoint quadrature over electrode facets.

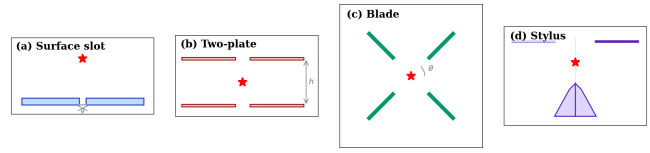

FIG. 1. Cross-sections of the four trap geometry families. (a) Surface slot: electrode width $W = 200\,\mu$m, gap $g = 10\,\mu$m, variable thickness t, ion height $d = 75\,\mu$m above top surface. (b) Two-plate: separation h, gap $g = 10\,\mu$m, ion at midplane. (c) Blade: four blades at $\pm\theta$, tip distance $R = 150\,\mu$m, thickness $b_t = 20\,\mu$m. (d) Stylus (axisymmetric): tip radius $r_{\rm tip}$, ring ground plane. Red stars mark the ion position.

Five Laplace solves are required per geometry. Mesh convergence is reported in Appendix A. Internal consistency is verified by symmetry checks: $\mathcal{G}_x = \mathcal{G}_y$ at $\theta = 45°$ to 0.003%, complementarity $\mathcal{G}_y(\theta) = \mathcal{G}_x(90° - \theta)$ to numerical precision, and $\mathcal{G}_x$ invariance ($< 1\%$) under the slot thickness sweep.

D. Geometry families

Figure 1 shows the four trap cross-sections studied in this work, chosen to span a range of enclosure conditions. (a) The *surface-electrode slot* is a planar geometry with a central gap of width g cut to variable depth t; the ion sits at height $d = 75\,\mu$m above the top surface. This geometry isolates the effect of the gap trench on the noise kernel. (b) The *two-plate trap* encloses the ion between parallel conducting plates separated by h, each containing a central gap; the ion sits at the midplane. (c) The *blade trap* consists of four blade electrodes arranged at angles $\pm\theta$ from the horizontal, with tip-to-center distance R. Varying θ continuously interpolates between horizontally and vertically dominated reflecting surfaces. (d) The *stylus trap* is an axisymmetric geometry consisting of a conducting needle (tip radius $r_{\rm tip}$) facing a ring ground plane. Unlike the other three cases, it is treated in the full 3D axisymmetric formulation and provides the clearest test of non-uniform distance scaling.

III. RESULTS

A. Noise recycling in the electrode gap

Figure 2 shows $\mathcal{G}_y$ as a function of the slot aspect ratio t/g. $\mathcal{G}_y$ increases by 30% from the flush geometry ($t/g = 0$) to $t/g = 4$, saturating at $t/g \approx 1$.

Decomposing $\mathcal{G}_y$ by surface region (Table I), the gap sidewalls contribute at most 2.2%. The enhancement is almost entirely on the *top surface*, concentrated within

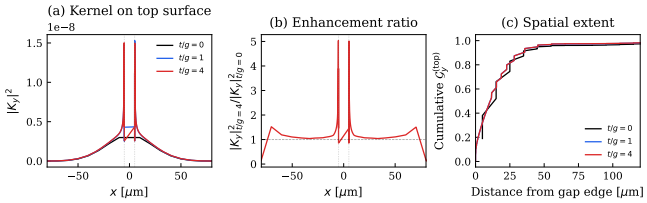

FIG. 2. Noise recycling in the slot (fixed $g = 10\,\mu$m, $d = 75\,\mu$m). (a) $|K_y|^2$ on the top surface for $t/g = 0, 1, 4$; the gap-edge spike appears only for finite thickness. (b) Enhancement ratio peaks at $\sim 5\times$ at the gap edge. (c) Cumulative $\mathcal{G}_y^{(\mathrm{top})}$: enhancement is localized within $\sim 2g$ of the gap edge.

$\sim 20\,\mu$m of the gap edge, where $|K_y|^2$ shows a $\sim 5\times$ spike relative to the flush case.

TABLE I. Slot sweep: $\mathcal{G}_y$ by surface region (fixed $g = 10\,\mu$m, $d = 75\,\mu$m).

t/g	$\mathcal{G}_y$ (total)	$\mathcal{G}_y$ (top surface)	Gap walls
0.00	1.57×10^{-7}	1.57×10^{-7}	0.0%
0.50	1.97×10^{-7}	1.94×10^{-7}	1.5%
1.00	2.00×10^{-7}	1.96×10^{-7}	2.2%
4.00	2.03×10^{-7}	1.99×10^{-7}	2.2%

The results support the noise recycling picture: in the finite-depth slot, image-path contributions that would escape through a flush opening are partially retained by the sidewalls and reappear as a localized enhancement near the gap edge. Saturation at $t/g \approx 1$ is explained by evanescent decay of Laplace solutions inside the slot (Appendix B). $\mathcal{G}_x$ varies by less than 1% across the sweep, consistent with the x-symmetry of the slot.

B. Enclosure enhancement in two-plate traps

Adding a second electrode plate above the ion increases $\mathcal{G}_y$ substantially in the 2D cross-section. Table II compares two-plate and surface trap values at matched d.

TABLE II. Two-plate vs. surface trap at matched $d = h/2$. Values are for the 2D cross-section; 3D corrections are expected to reduce the ratio (see text).

h [μm]	d	$\mathcal{G}_y d^3$ (2-plate)	$\mathcal{G}_y d^3$ (surface)	Ratio
200	100	0.231	0.088	2.6×
400	200	0.244	0.096	2.5×

The $\sim 2.5\times$ enhancement in the 2D cross-section is consistent with the addition of bouncing-ball ray paths between the two plates which are absent in the open surface geometry. In a finite-length 3D trap, end effects will reduce the enclosure; we expect the enhancement to persist qualitatively but with smaller magnitude.

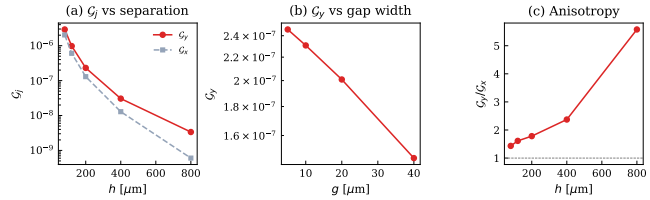

FIG. 3. Two-plate results (fixed $W = 200\,\mu$m). (a) $\mathcal{G}_j$ vs. plate separation h (gap $g = 10\,\mu$m). (b) $\mathcal{G}_y$ vs. gap width g ($h = 200\,\mu$m). (c) Anisotropy $\mathcal{G}_y/\mathcal{G}_x$ vs. h.

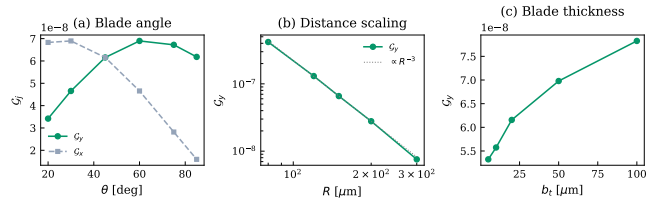

FIG. 4. Blade trap results. (a) $\mathcal{G}_x$, $\mathcal{G}_y$ vs. blade angle θ (fixed $R = 150\,\mu$m, $b_t = 20\,\mu$m); exact crossing at 45°. (b) Distance scaling $\mathcal{G}_y$ vs. R at $\theta = 45°$, with R^{-3} reference. (c) $\mathcal{G}_y$ vs. blade thickness b_t ($R = 150\,\mu$m, $\theta = 45°$).

The anisotropy $\mathcal{G}_y/\mathcal{G}_x$ grows from 1.4 at $h = 80\,\mu$m to 5.6 at $h = 800\,\mu$m: vertical recycling paths preferentially enhance $\mathcal{G}_y$. Wider gaps reduce the enhancement: $\mathcal{G}_y$ drops 41% from $g = 5\,\mu$m to $g = 40\,\mu$m at fixed $h = 200\,\mu$m, as more ray paths escape.

C. Blade angle controls heating anisotropy

The blade angle θ controls which motional mode couples more strongly to surface noise. Figure 4 shows $\mathcal{G}_x$ and $\mathcal{G}_y$ vs. θ.

At $\theta = 45°$, four-fold symmetry gives $\mathcal{G}_x = \mathcal{G}_y$, satisfied to 0.003% numerically. The anisotropy ratio spans $\mathcal{G}_y/\mathcal{G}_x = 0.50$ ($\theta = 20°$) to 3.88 ($\theta = 85°$), with $\mathcal{G}_y(\theta) = \mathcal{G}_x(90° - \theta)$ confirmed to numerical precision. The ratio is robust under mesh refinement: at $\theta = 85°$, the anisotropy ratio converges to within 3% across a 4× range of mesh sizes, even though the absolute $\mathcal{G}_y$ varies by $\sim 14\%$ (Appendix A).

The 2D distance scaling $\mathcal{G}_y \propto R^{-3}$ holds with $\mathcal{G}_y \cdot R^3 = 0.218 \pm 0.008$ across $R \in [80, 300]\,\mu$m. Blade thickness also has a moderate effect: $\mathcal{G}_y$ increases 47% from $b_t = 5\,\mu$m to $100\,\mu$m at fixed R and θ, because thicker blades present more reflecting area.

D. Deviation from d^{-4} in the stylus trap

The stylus trap illustrates the distinction between uniform scaling and ion-position sweeps within a fixed geometry, previewed in the Introduction. Under *uniform* scaling, $\mathcal{G}_j \propto d^{-4}$ (3D) or d^{-3} (2D) is exact [3]. When

the ion moves within a *fixed* geometry, the exponent depends on how the geometric structure visible from the ion changes with position.

If the scaling were d^{-4}, $\mathcal{G}_z \cdot z_{\rm ion}^4$ would be constant; instead it varies by 3.8× (Table III). A fit to $\log \mathcal{G}_z$ vs. $\log z_{\rm ion}$ gives $\alpha_{\rm eff} = -4.69 \pm 0.15$. The deviation is physical: at small $z_{\rm ion}$ the stylus cap subtends a large solid angle and dominates $\mathcal{G}_z$ (Table IV); as the ion recedes, both the distance and the subtended solid angle decrease, steepening the falloff beyond d^{-4}. In contrast, a symmetric geometry (e.g., the two-plate trap at the midplane) should exhibit deviations closer to zero.

IV. DISCUSSION

A. Noise recycling as an organizing interpretation

Across all four geometries, $\mathcal{G}_j$ increases when reflecting electrode surfaces retain image paths that would otherwise escape.

The surface slot adds recycling within the trench (+30% in $\mathcal{G}_y$, localized at the gap edge). The blade trap creates four isolated reflectors ($\mathcal{G}_y \cdot d^3 \approx 0.22$ in 2D). The two-plate trap adds bouncing-ball orbits ($\mathcal{G}_y \cdot d^3 \approx 0.23$ in 2D, vs. 0.08 for the open surface). The stylus tip concentrates the reflecting surface in a small solid angle, producing an effective scaling exponent steeper than d^{-4}.

We emphasize that noise recycling is a geometric interpretation of the numerically computed $\mathcal{G}_j$, not a substitute for the Green's-function calculation and not a rigorous derivation. Its value is organizational: it explains why enhancement localizes near the slot edge, why enclosure raises $\mathcal{G}_y$, and why blade orientation controls anisotropy, while the quantitative magnitudes remain those of the numerical Green's-function solution.

B. Design implications

Enclosure carries a noise cost. In the 2D cross-section, traps with reflective surfaces above the ion (package lids, ground-plane shields) pay a $\sim$2–3× penalty in $\mathcal{G}_y$. In a 3D device, end effects and finite electrode length will moderate this factor, but the qualitative direction—more enclosure increases $\mathcal{G}_y$—should persist.

TABLE III. Stylus: $\mathcal{G}_z$ vs. ion height (fixed domain: $h_{\rm ground} = 500\,\mu$m, $r_{\rm tip} = 10\,\mu$m).

$z_{\rm ion}$ [µm]	$\mathcal{G}_z$	$\mathcal{G}_z \cdot z_{\rm ion}^4$
30	3.05×10^{-7}	0.247
50	1.84×10^{-8}	0.115
80	1.92×10^{-9}	0.079
120	3.15×10^{-10}	0.065
200	4.30×10^{-11}	0.069

Blade angle predicts which mode heats faster. In blade traps, the two radial heating rates can be compared as a function of blade angle θ. For nearly degenerate radial modes, or after correcting for the frequency dependence in Eq. (4), the measured heating-rate anisotropy should track $\mathcal{G}_y/\mathcal{G}_x$ as a function of θ. The two modes should heat equally at $\theta = 45°$; more generally, Eq. (4) gives the appropriate frequency correction. Deviations from the predicted trend serve as a consistency check on the patch-potential model.

Distance-scaling exponents are geometry-specific. Experiments that vary the ion–electrode distance within a fixed trap [6–8] should observe an effective exponent $\alpha_{\rm eff}$ that differs from -4 by an amount computable from the geometry. The asymmetric stylus shows the largest deviation ($\alpha_{\rm eff} \approx -4.7$); symmetric enclosed traps should show the smallest.

C. Relation to other noise mechanisms

The geometry functional $\mathcal{G}_j$ enters identically for any noise model that produces spatially delta-correlated voltage fluctuations on the electrode surface, including thermally activated adatom diffusion [11] and thermomechanical fluctuations. The noise recycling enhancement is therefore not specific to the patch-potential picture: any surface noise source whose spectrum factorizes as in Eq. (2) will inherit the same geometry-dependent amplification. For noise models with finite spatial correlation length ℓ, the geometry functional generalizes to a double surface integral weighted by the correlation function; our results apply in the limit $\ell \ll g$.

D. Limitations

The blade and two-plate computations are 2D cross-sections, assuming translational invariance along the trap axis. For finite blade length, 3D corrections will reduce the effective recycling; we expect the *trends*—anisotropy controlled by blade angle, enclosure enhancing $\mathcal{G}_y$—to persist with reduced magnitudes, but the precise numerical factors (the 2.5× ratio, the 0.5–3.9 anisotropy range) should be taken as cross-sectional predictions, not as exact 3D values. The stylus uses the full 3D axisymmetric formulation.

All results assume spatially delta-correlated noise ($\ell \to 0$). At finite correlation length $\ell \gtrsim g$, the gap-edge ker-

TABLE IV. Stylus: $\mathcal{G}_z$ decomposition by surface region ($z_{\rm ion} = 80\,\mu$m).

$r_{\rm tip}$ [µm]	$\mathcal{G}_z$	Cap	Wall	Ground
2	1.42×10^{-9}	71%	15%	14%
10	2.28×10^{-9}	87%	6%	7%
25	6.04×10^{-9}	97%	<1%	3%

nel spike would be smoothed and the slot enhancement reduced.

Mesh convergence (Appendix A) shows that absolute values of $\mathcal{G}_j$ are converged to within $\sim 1\%$ for two-plate and stylus geometries, but have $\sim 14\%$ mesh sensitivity for the blade at $\theta = 85°$. The *anisotropy ratios* are more robust, converging to within 3% for all cases tested.

V. CONCLUSION

We have computed the geometry functional $\mathcal{G}_j$ for four families of trap cross-sections and shown that their main trends are coherently organized by a noise-recycling picture, in which reflecting boundaries retain electrostatic image paths that would otherwise escape. Three predictions follow:

1. Two traps with different enclosure but matched d, comparable surface preparation, and either matched secular frequency or an Eq. (4) correction should exhibit heating-rate ratios computable from $\mathcal{G}_j$ (a factor of ~ 2.5 in the 2D cross-section for two-plate vs. open surface).

2. In blade traps, the geometry ratio $\mathcal{G}_y/\mathcal{G}_x$ is a function of the blade angle θ, with an exact isotropic point at $\theta = 45°$; for nearly degenerate radial modes, the measured heating-rate anisotropy should follow the same trend.

3. Distance-scaling measurements in asymmetric traps (e.g., stylus) will show effective exponents deviating from d^{-4}, computable from the geometry.

Extensions to full 3D geometries and finite noise correlation length are natural next steps. Numerical code and data are available at `https://github.com/lostree9/noise_recycling`.

ACKNOWLEDGMENTS

This work was completed as part of an undergraduate thesis at the University of Massachusetts Amherst under the supervision of C. Cox and R. Niffenegger. Computations were performed on the Unity high-performance computing cluster.

Appendix A: Convergence tests

Table V reports $\mathcal{G}_j$ for three geometries across a range of mesh sizes, spanning a 4× refinement factor. The mesh size parameter h_{mesh} controls the characteristic element length near electrode surfaces; refinement near the ion and electrode tips is set proportionally.

The two-plate geometry is converged to better than 0.1%. The stylus is converged to $\sim 1.4\%$. The blade at $\theta = 85°$ shows $\sim 14\%$ variation in $\mathcal{G}_y$ between the finest and middle meshes, attributable to the sharp kernel spike at the blade tips. However, the anisotropy ratio $\mathcal{G}_y/\mathcal{G}_x$ converges to within 3%, because mesh errors affect both components similarly. All trends and comparisons reported in the main text are robust across this range.

TABLE V. Mesh convergence. h_{mesh} is the mesh size factor (smaller = finer). For the blade, the anisotropy ratio $\mathcal{G}_y/\mathcal{G}_x$ converges faster than the absolute value.

Case h_{mesh}	$\mathcal{G}_y$	$\mathcal{G}_x$	$\mathcal{G}_y/\mathcal{G}_x$
Blade, $\theta = 85°$, $R = 150\,\mu$m			
0.75	7.07×10^{-8}	1.77×10^{-8}	3.98
1.5	6.18×10^{-8}	1.60×10^{-8}	3.88
3.0	5.23×10^{-8}	1.35×10^{-8}	3.87
Two-plate, $h = 800\,\mu$m, $g = 10\,\mu$m			
2.0	3.365×10^{-9}	—	—
4.0	3.363×10^{-9}	—	—
8.0	3.366×10^{-9}	—	—
Stylus, $z_{\text{ion}} = 200\,\mu$m, $r_{\text{tip}} = 10\,\mu$m			
0.75	4.328×10^{-11}	—	—
1.5	4.302×10^{-11}	—	—
3.0	4.268×10^{-11}	—	—

Appendix B: Evanescent model for slot saturation

The saturation of $\mathcal{G}_y$ at $t/g \approx 1$ can be understood from the evanescent structure of Laplace solutions inside the slot.

1. Waveguide modes in the slot

The slot is a channel of width g bounded by conducting sidewalls. Solutions of $\nabla^2\varphi = 0$ inside the slot that vanish on both sidewalls take the form

$$\varphi_n(x, y) = \sin\left(\frac{n\pi x}{g}\right) e^{\pm n\pi y/g}, \quad n = 1, 2, 3, \ldots \tag{B1}$$

These are evanescent: the lowest mode ($n = 1$) decays with characteristic length g/π. A field perturbation entering the slot from the top decays by a factor $e^{-\pi y/g}$ at depth y.

2. Round-trip attenuation

A perturbation that enters the slot, propagates to the conducting bottom at depth t, reflects, and returns to the top surface is attenuated by the round-trip factor

$$\eta(t/g) = e^{-2\pi t/g}. \tag{B2}$$

This controls how much the slot bottom contributes to the Green's function at the top surface. For $t/g = 0.5$:

$\eta \approx 0.04$. For $t/g = 1$: $\eta \approx 0.002$. For $t/g \geq 1$, the slot bottom is effectively invisible, and adding further depth changes $\mathcal{G}_y$ negligibly.

3. Image-sum interpretation

Equivalently, in the method-of-images picture, the slot bottom at depth t generates a sequence of images at depths $2nt$ ($n = 1, 2, \ldots$), each contributing to the kernel on the top surface. For a point at distance x' from the gap edge, the contribution of the n-th image to $\partial G/\partial n$ scales as

$$\delta K \sim \frac{(-1)^n}{\sqrt{(2nt)^2 + x'^2}}. \tag{B3}$$

The alternating sign arises from the Dirichlet condition (sign flip at each reflection). Integrating $|\delta K|^2$ over the near-gap region (x' from 0 to $\sim g$), each image contributes a term proportional to a decreasing function $f(2nt/g)$ with $f(u) \to 0$ as $u \to \infty$. The total enhancement is an alternating series

$$\frac{\mathcal{G}_y(t)}{\mathcal{G}_y(0)} - 1 \sim \sum_{n=1}^{\infty} (-1)^{n+1} f\left(\frac{2nt}{g}\right), \tag{B4}$$

which converges rapidly once the first term ($n = 1$, argument $2t/g$) becomes small—i.e., when $t/g \gtrsim 1$.

4. Comparison with data

The evanescent model predicts an enhancement of the form

$$\frac{\mathcal{G}_y(t)}{\mathcal{G}_y(0)} = 1 + C_\infty\big(1 - e^{-2\pi t/g}\big), \tag{B5}$$

where C_∞ is the maximum fractional enhancement, set by the fraction of $\mathcal{G}_y$ sensitive to the gap region. A one-parameter fit to the data gives $C_\infty = 0.30$:

TABLE VI. Evanescent model vs. numerical data.

t/g	$\mathcal{G}_y/\mathcal{G}_y(0)$ (data)	Eq. (B5)	Error
0.25	1.196	1.24	3.4%
0.50	1.255	1.29	2.5%
1.00	1.277	1.30	1.8%
4.00	1.297	1.30	0.2%

The simple model overestimates the rate of saturation (predicting convergence slightly faster than observed), consistent with the fact that the slot opening is not a perfect conducting boundary: the open mouth couples to the half-space above, reducing the effective decay rate below the ideal π/g. Nevertheless, the qualitative prediction—saturation at $t/g \sim O(1)$, independent of d—is confirmed.

[1] Q. A. Turchette, D. Kielpinski, B. E. King, D. Leibfried, D. M. Meekhof, C. J. Myatt, M. A. Rowe, C. A. Sackett, C. S. Wood, W. M. Itano, C. Monroe, and D. J. Wineland, Heating of trapped ions from the quantum ground state, Phys. Rev. A **61**, 063418 (2000).

[2] M. Brownnutt, M. Kumph, P. Rabl, and R. Blatt, Ion-trap measurements of electric-field noise near surfaces, Rev. Mod. Phys. **87**, 1419 (2015).

[3] G. H. Low, P. F. Herskind, and I. L. Chuang, Finite-geometry models of electric field noise from patch potentials in ion traps, Phys. Rev. A **84**, 053425 (2011).

[4] K.-Y. Lin, G. H. Low, and I. L. Chuang, Effects of electrode surface roughness on motional heating of trapped ions, Phys. Rev. A **94**, 013418 (2016).

[5] DOLFINx developers, DOLFINx: Next generation FEniCS problem solving environment (2023), `https://github.com/FEniCS/dolfinx`.

[6] I. A. Boldin, A. Kraft, and C. Wunderlich, Measuring anomalous heating in a planar ion trap with variable ion-surface separation, Phys. Rev. Lett. **120**, 023201 (2018).

[7] J. A. Sedlacek, A. Greene, J. Stuart, R. McConnell, C. D. Bruzewicz, J. M. Sage, and J. Chiaverini, Distance scaling of electric-field noise in a surface-electrode ion trap, Phys. Rev. A **97**, 020302(R) (2018).

[8] L. Deslauriers, S. Olmschenk, D. Stick, W. K. Hensinger, J. Sterk, and C. Monroe, Scaling and suppression of anomalous heating in ion traps, Phys. Rev. Lett. **97**, 103007 (2006).

[9] D. A. Hite, Y. Colombe, A. C. Wilson, K. R. Brown, U. Warring, R. Jördens, J. D. Jost, K. S. McKay, D. P. Pappas, D. Leibfried, and D. J. Wineland, 100-fold reduction of electric-field noise in an ion trap cleaned with *in situ* argon-ion-beam bombardment, Phys. Rev. Lett. **109**, 103001 (2012).

[10] J. Chiaverini and J. M. Sage, Insensitivity of the rate of ion motional heating to trap-electrode material over a large temperature range, Phys. Rev. A **89**, 012318 (2014).

[11] A. Safavi-Naini, P. Rabl, P. F. Weck, and H. R. Sadeghpour, Microscopic model of electric-field-noise heating in ion traps, Phys. Rev. A **84**, 023412 (2011).

[12] C. D. Bruzewicz, J. Chiaverini, R. McConnell, and J. M. Sage, Trapped-ion quantum computing: Progress and challenges, Appl. Phys. Rev. **6**, 021314 (2019).